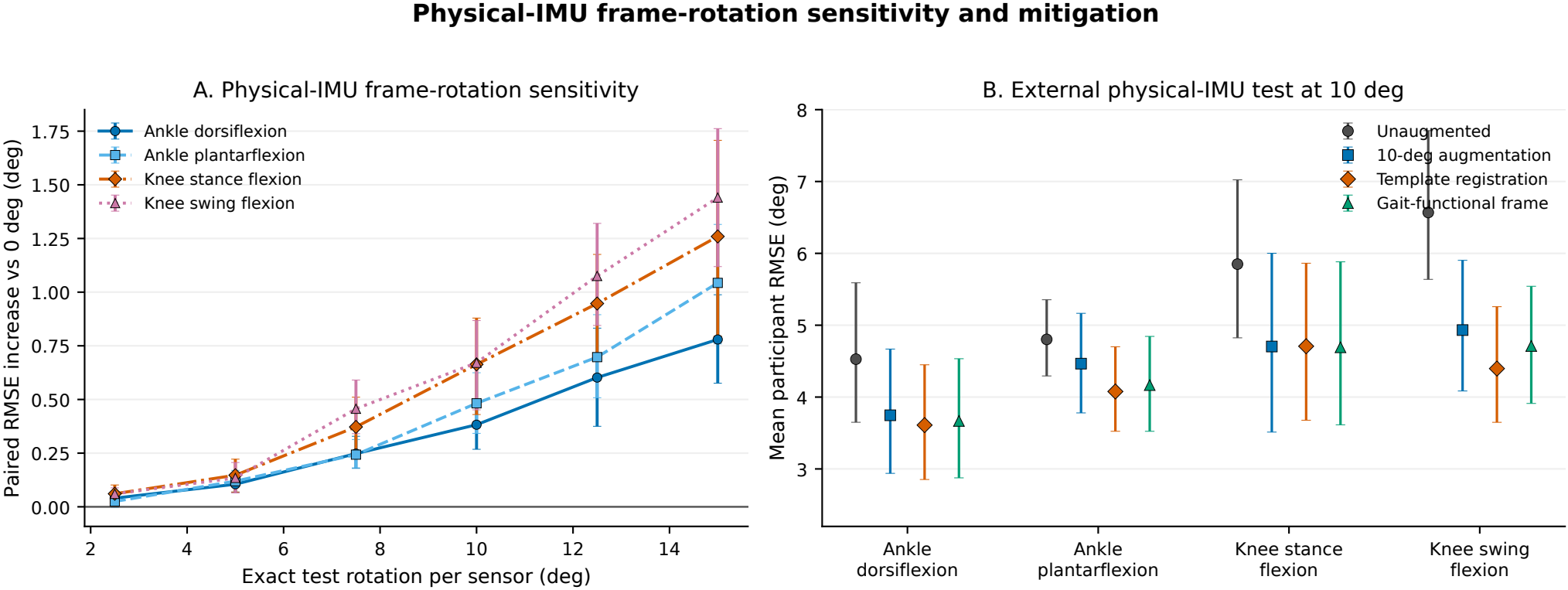
Physical-IMU frame-rotation sensitivity and mitigation
A. Physical-IMU frame-rotation sensitivity
Ankle dorsiflexion
Ankle plantarflexion
Knee stance flexion
Knee swing flexion
Paired RMSE increase vs 0 deg (deg)
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2
4
6
8
10
12
14
Exact test rotation per sensor (deg)
B. External physical-IMU test at 10 deg
Unaugmented
10-deg augmentation
Template registration
Gait-functional frame
Mean participant RMSE (deg)
3
4
5
6
7
8
Ankle dorsiflexion
Ankle plantarflexion
Knee stance flexion
Knee swing flexion